

PhotonGIS User Guide

1. What is PhotonGIS?

PhotonGIS is a solar photovoltaic (PV) planning tool for designing solar panel installations on roofs.

Users can:

- Create and manage PV projects.
 - Add custom solar panels, including their physical dimensions and electrical properties.
 - Add custom inverters and configure their electrical parameters.
 - Locate a roof using a satellite map.
 - Place and arrange solar panels directly on the roof.
 - Group panels according to their roof slope and orientation.
 - Connect panels into inverter strings.
 - Analyze expected monthly and yearly solar power production.
 - Analyze production and electrical characteristics for individual panel groups, inverters, MPPTs, and strings.
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2. Projects

Before placing panels, the user must create or open a project.

A newly created project is initially empty and does not contain any panels.

If the user is new to PhotonGIS, they may also need to create their solar panels and inverters.

Panels and inverters can be added in two ways:

- From the main dashboard.
 - Directly from the project/map editor while designing a project.
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3. Finding the Roof Location

Before placing panels, the user must navigate to the desired roof location.

There are two ways to find a location:

Using the satellite map

The user can navigate manually through the satellite map until the desired roof is visible.

Using address search

The user can use the search function, represented by the magnifying-glass icon in the top-right corner of the screen.

Enter the desired address to navigate to that location.

4. Placing Solar Panels

4.1 Panel groups

Panels should be grouped according to their physical roof characteristics.

Panels belonging to the same panel group must have:

- The same roof slope (tilt angle).
- The same panel orientation.

If panels have different slopes or orientations, they should be placed in separate panel groups.

4.2 Start panel placement

To start placing panels:

1. Open the panel-placement tool by clicking the panel-and-sun icon in the bottom menu. This is the first button in the bottom menu.
2. Select an existing panel from the panel dropdown.
3. If no suitable panel exists, select **"Add a new panel"** and create the panel first.
4. After creating the panel, select it from the dropdown.

4.3 Configure the panel group

Before placing the panels, configure the panel group.

Roof slope

Enter the roof slope in degrees in the slope field in the top-left corner of the panel placement form.

The slope represents the angle of the roof.

Panel orientation

Select the panel orientation on the right side of the panel placement form.

Available orientations:

- Vertical
- Horizontal

Vertical is selected by default.

Panel spacing

The bottom input controls the spacing between panels and between rows.

Enter the spacing in millimeters.

4.4 Place the first panel

After configuring the panel group, click on the desired location on the roof to place the first panel.

The first panel becomes the initial/reference panel for the group.

After the first panel is placed, plus (+) icons appear around it.

These icons allow the user to add additional panels to the corresponding side of the initial panel.

4.5 Move a panel group

While the panel group is still in editing mode, the entire group can be moved.

Drag the initial panel to move the group across the map.

The initial panel is displayed in blue while the group is being edited.

4.6 Change panel orientation

While editing a panel group, a panel-orientation slider appears at the top of the screen.

Use this slider to adjust the orientation of the panel group.

4.7 Save a panel group

When the panel group is configured correctly, click "**Save**" at the top of the screen.

The group is then saved to the project.

5. Editing Existing Panel Groups

To edit an existing panel group:

1. Click any panel belonging to the group.
 2. The panel group enters editing mode.
 3. Modify the group as required.
 4. Save the changes.
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6. Deleting Panels

Delete an individual panel

To delete an individual panel:

1. Click the green panel.
2. Delete the panel using the available deletion control.

The initial/reference panel cannot be deleted individually.

Delete an entire panel group

To delete the entire panel group:

1. Select the panel group.
2. Enter editing mode.
3. Click "**Delete**" at the top of the screen next to the orientation slider.

Important deletion rule

A panel that is connected to an inverter string cannot be deleted.

If a panel belongs to an inverter string, the user must first delete the corresponding inverter string before the panel can be deleted.

7. Solar Production Analytics

To view project analytics, click the thunderbolt icon in the bottom menu.

A popup containing solar production analytics will open.

7.1 Project-level analytics

The first section displays analytics for the entire currently opened project.

It includes:

- Monthly solar production.
- Yearly solar production.
- Installed system power in kW.
- Yearly production in kWh.
- Total number of installed panels.

7.2 Panel-group analytics

Below the project-level analytics are analytics for individual panel groups.

Panel groups may have different:

- Roof slopes.
- Orientations.
- Panel types.
- Production characteristics.

Each panel group therefore has its own analytics.

A panel-group analytics section includes:

- Monthly production chart.
- Yearly production chart.
- Installed power in kW.
- Yearly production in kWh.
- Roof slope.
- Number of panels.
- Panel details.

A panel group can be deleted from its analytics section by clicking the trash-can icon on the right side of the group.

8. Inverters and Strings

Panels can be connected to inverters through strings.

To start configuring an inverter:

1. Click the diagram icon in the bottom menu. The icon resembles three connected dots.
2. A popup opens.

3. Select an existing inverter from the dropdown.
4. If the required inverter does not exist, select "**Add a new inverter**".
5. Enter the inverter's parameters.
6. After creating the inverter, select it from the dropdown.

9. Creating an Inverter String

After selecting an inverter, click "**Design string**" to start creating the first string.

To add panels to a string:

1. Click the first panel.
2. Continue clicking panels that should belong to the string.
3. PhotonGIS checks the inverter limits while panels are being selected.

The available inverter limits are displayed in the helper panel on the right side of the screen.

10. Inverter Limits

Inverter limits are determined by the parameters configured for the inverter.

Limits are evaluated per string and per MPPT and include electrical limits such as:

- Current.
- Voltage.

When adding panels would cause the string to exceed an applicable inverter limit, those panels become unavailable and are displayed in gray.

The user cannot select panels if doing so would exceed the applicable inverter limits.

Minimum MPPT voltage

A string must reach at least the configured MPPT minimum voltage before it can be saved.

If the minimum MPPT voltage has not been reached, the user cannot save the string.

11. Saving and Managing a String

After selecting the desired panels, press the **Enter** key.

Additional controls appear at the top of the screen next to **"Cancel"**.

Cancel

"Cancel" cancels the current string-creation process and exits the string editor.

Save

"Save" saves the currently configured string.

The string must satisfy the applicable inverter/MPPT requirements before it can be saved.

Continue

"Continue" allows the user to continue the string-design workflow.

12. Removing the Last Panel From a String

While creating or editing a string, press the **Z** key to disconnect the most recently added panel.

The user can press **Z** multiple times to remove multiple panels, working backward through the string until the first connected panel is reached.

13. Inverter Analysis

The inverter/string management interface can also be accessed from the left sidebar.

The same three-connected-circles icon used to start inverter configuration is available in the sidebar.

This section displays all inverters currently used in the project.

13.1 Inverter analytics

Click an inverter to open its analytics.

The inverter section includes a yearly solar production chart for that inverter.

Below the chart are the inverter's MPPTs.

An inverter may have multiple MPPTs depending on its configuration.

13.2 MPPT and string management

Select an MPPT to view the strings connected to it.

Strings are listed below the selected MPPT.

Click a string to open its detailed analytics.

String analytics include:

- Yearly solar production.
- String voltage.
- String current.

13.3 Delete a string

To delete a string:

1. Select the string.
2. Click the trash-can icon in the top-left corner of the string section.

After deleting a string, a new string can be created using the **"Add string"** button below the MPPT.

If the MPPT supports multiple strings, **"Add string"** remains available even when another string is already connected to that MPPT.

14. Delete an Inverter

To delete an inverter:

1. Open the inverter analysis section.
2. Scroll to the bottom.
3. Click the trash-can icon.

Deleting an inverter removes the selected inverter from the project.

Before deleting an inverter, ensure that its connected strings are handled appropriately.

15. Important Product Rules

The following rules should always be respected:

- Panels with different roof slopes should belong to different panel groups.
- Panels with different orientations should belong to different panel groups.
- The initial/reference panel cannot be deleted individually.
- A panel connected to an inverter string cannot be deleted until the corresponding string is deleted.
- A string cannot be saved until it reaches the MPPT minimum voltage.
- Panels cannot be added to a string when doing so would exceed the applicable inverter current or voltage limits.
- Inverter limits are evaluated per string and per MPPT.
- MPPTs may support different numbers of strings depending on the inverter configuration.
- Project-level analytics represent the entire currently opened project.
- Panel-group analytics represent only the selected panel group.
- Inverter analytics represent the selected inverter.
- String analytics represent the selected string.

16. Adding a New Panel

Panels can be created from the dashboard or directly from the panel-placement workflow.

When creating a new panel, enter its basic information, physical dimensions, and electrical specifications.

16.1 Basic Information

Panel name

Enter a name for the panel.

The panel name is used to identify the panel when selecting it for a project or panel group.

16.2 Dimensions

Enter the physical dimensions of the panel.

Height

Enter the panel height in millimeters (mm).

Width

Enter the panel width in millimeters (mm).

The dimensions determine the physical size of the panel when it is placed on the satellite map.

16.3 Electrical Specifications

Enter the electrical characteristics of the panel.

Power

Enter the panel's rated power in watts (W).

V_{mp}

Enter the panel's voltage at maximum power in volts (V).

V_{mp} is the voltage at which the panel produces its maximum rated power.

V_{oc}

Enter the panel's open-circuit voltage in volts (V).

V_{oc} is the panel voltage when no current is being drawn from the panel.

I_{sc}

Enter the panel's short-circuit current in amperes (A).

I_{sc} is the maximum current produced by the panel under the specified test conditions when the circuit is shorted.

I_{mp}

Enter the panel's current at maximum power in amperes (A).

I_{mp} is the current produced by the panel at its maximum power point.

These electrical specifications are used when evaluating panel strings and determining whether a string is within the connected inverter's electrical limits.

17. Adding a New Inverter

Inverters can be created from the dashboard or directly from the inverter/string configuration workflow.

When creating an inverter, enter its technical specifications.

17.1 Basic Information

Name

Enter a name for the inverter.

The inverter name is used to identify it when selecting an inverter for a project.

17.2 Power

The inverter power specifications describe the electrical power limits of the inverter.

AC Power

Enter the inverter's AC power rating in watts (W).

DC Power

Enter the inverter's DC power rating in watts (W).

17.3 DC Voltage Range

Enter the inverter's applicable DC voltage limits.

Max DC

Enter the maximum supported DC voltage in volts (V).

A panel string must not exceed the applicable maximum DC voltage.

Start

Enter the inverter's startup voltage in volts (V).

This represents the DC voltage required for the inverter to start operating.

MPPT Min

Enter the minimum voltage required by the MPPT in volts (V).

A string must reach at least this voltage before the string can be saved.

MPPT Max

Enter the maximum voltage supported by the MPPT in volts (V).

A string must not exceed this MPPT voltage limit.

17.4 MPPT Configuration

Count

Enter the number of MPPTs available on the inverter.

Each MPPT can have its own connected strings and electrical limits.

The number of MPPTs determines how many MPPT sections are available when configuring the inverter.

Current

Enter the maximum supported current in amperes (A).

The applicable string/MPPT current must remain within this limit.

Strings (per MPPT)

Enter the maximum number of strings that can be connected to each MPPT.

This determines whether multiple strings can be connected to the same MPPT.

For example, if **Strings (per MPPT)** is 2, an MPPT can have up to two connected strings.

17.5 Efficiency

Enter the inverter's efficiency.

Efficiency represents the percentage of DC power that is converted into usable AC power.

The configured efficiency is used when calculating the expected power production of the system.

18. Panel and Inverter Data in String Design

The electrical specifications entered for panels and inverters are important because PhotonGIS uses them when designing and validating strings.

For panels, the relevant electrical parameters include:

- Power.
- V_{mp} .
- V_{oc} .
- I_{mp} .
- I_{sc} .

For inverters, the relevant parameters include:

- DC power.
- Maximum DC voltage.
- Startup voltage.
- MPPT minimum voltage.
- MPPT maximum voltage.
- MPPT current.
- Number of MPPTs.
- Maximum strings per MPPT.
- Efficiency.

When panels are connected into a string, PhotonGIS uses these parameters to determine whether the resulting string is compatible with the selected inverter and MPPT.

If adding a panel would exceed an applicable inverter or MPPT limit, the panel cannot be selected.

A string also cannot be saved until it satisfies the MPPT minimum-voltage requirement.